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ORIGINAL ARTICLE

A reference genome and transcriptome of haustorial development in *Pedicularis groenlandica* reveal diverse trajectories of haustoria-associated gene evolution in parasitic plants

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• **Background and Aims** Novel traits frequently evolve by co-opting existing genetic pathways through either the direct repurposing of existing genes or neofunctionalization of duplicated genes. Parasitism represents a major innovation in plants, in which a novel organ, the haustorium, evolved to penetrate hosts to extract water and nutrients. Previous studies hypothesized that haustoria-associated genes evolved primarily from root and pollen-associated pathways by neofunctionalization following genome duplications.
• **Methods** To examine the evolutionary origins of haustoria-associated genes, we generated a chromosome-scale reference genome of the hemiparasite *Pedicularis groenlandica* (Orobanchaceae) and sampled transcriptomes throughout haustorial development. We identified haustoria-associated genes; quantified the expression of these genes and their paralogues across non-haustorial tissues; and investigated overlap in haustoria-associated genes with publicly available haustorial transcriptomes in four additional parasitic plants.
• **Key Results** We identified 5635 haustoria-associated genes in *P. groenlandica*. A greater proportion of these genes were associated with pollen tubes (67 %) than with roots (33 %), evidenced by being either differentially expressed in pollen tubes or nested within pollen tube-associated gene families, or both. Haustoria-associated genes with paralogues that arose after the evolution of parasitism in Orobanchaceae are more likely to be uniquely expressed in haustoria, consistent with neofunctionalization.
• **Conclusions** Our results support evidence of both pleiotropy and neofunctionalization as mechanisms by which genetic pathways are co-opted for haustorial function. Haustoria-associated genes are highly lineage specific, highlighting a dynamic and ongoing process of independent haustorial co-option of gene functions among parasitic plant lineages.

Key words: Genome, haustoria, neofunctionalization, parasitic plants, pleiotropy, pollen tubes, transcriptomics.

INTRODUCTION

Approximately 1 % of flowering plants are parasites on other plants, a strategy that has evolved independently around a dozen times in angiosperms (Westwood *et al.*, 2010). Parasitic plants use a novel organ, the haustorium, to invade host tissues and establish a vascular connection to extract water and/or nutrients. The repeated convergent evolution of haustoria, including their origination from both root tissue tissue (e.g. *Striga*) and stem tissue (e.g. *Cuscuta*) in different lineages, raises fundamental questions about the genomic, developmental and ecological precursors that enable this transition. Comparative studies have revealed extensive anatomical and physiological convergence in haustoria across a range of lineages (Fernández-Aparicio *et al.*, 2016; Yoshida *et al.*, 2016; Masumoto *et al.*, 2021), and recent genomic studies have identified both lineage-specific and shared molecular pathways associated with haustorial development and function (Westwood *et al.*, 2012; Zhang *et al.*, 2015; Kokla *et al.*, 2022; Bawin *et al.*, 2022, 2023; Jhu *et al.*, 2022; Zou *et al.*, 2026). Despite our expanding knowledge of the genes that govern

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haustorial attachment across parasitic lineages, the evolutionary processes by which genes have been co-opted for haustorial function remain poorly understood.

Parasitic plants span a continuum of host dependence based on developmental requirements (facultative versus obligate parasites) and trophic capacity (hemiparasites, which retain photosynthetic capacity, versus holoparasites, which are fully heterotrophic). Facultative hemiparasites produce lateral haustoria that connect primarily to host xylem, whereas obligate parasites form a terminal haustorium that connects to both xylem and phloem (Press and Graves, 1995; Yoshida *et al.*, 2016). It has been hypothesized that plant parasitism evolves in a stepwise fashion, with facultative hemiparasitism representing an initial stage that gives rise to obligate hemiparasites and, ultimately, holoparasites (Westwood *et al.*, 2010; Xu *et al.*, 2022). The family Orobanchaceae uniquely spans this full spectrum and encompasses economically significant weeds, such as witchweeds (*Striga*) and broomrapes (*Orobanche*), that cause extensive damage to crops worldwide (Joel *et al.*, 2013; Li *et al.*, 2019). As a result, the Orobanchaceae have emerged as a model system for the study of haustorial physiology and the genomic basis of parasitic plant evolution.

Haustoria appear, at least in part, to co-opt genetic pathways active in other tissues. This reflects a general trend whereby plants redeploy existing gene repertoires to acquire novel functions and life-history strategies. For example, MADS-box genes, which functioned in haploid-to-diploid phase transitions in algae, are hypothesized to have been co-opted for floral organ development in angiosperms (Tanabe *et al.*, 2005), and terpene synthases, which functioned primarily in metabolic processes, were co-opted for specialized volatile production and herbivore defence (Abbas *et al.*, 2017). As root parasites, Orobanchaceae are hypothesized predominantly to co-opt root-associated genetic pathways for haustorial function, where orthologues of haustoria-associated genes are highly expressed in the roots of non-parasitic relatives (Yang *et al.*, 2015). Curiously, Yang *et al.* (2015) found that haustorial orthologues are often differentially expressed in several other tissues, including pollen and flowers, a result replicated by Yoshida *et al.* (2019). We hypothesize that this relationship between haustoria- and pollen-associated genes results from shared functions between haustoria and pollen tubes, including signal detection, polarized tip growth, cell wall remodelling and traversal of foreign tissues. The discovery of haustoria-associated gene homologues expressed in other tissue types suggests that neofunctionalization of paralogues might play a significant role in haustorial function. Alternatively, these genes might participate pleiotropically in haustorial function while also retaining function in other tissues through tissue-specific regulation.

The prevailing hypothesis for how plants transition to parasitism is that novel genes required for haustorial development arise after whole-genome duplications (WGDs) create duplicate gene copies that are co-opted and neofunctionalized (Vogel *et al.*, 2018; Lyko and Wicke, 2021). Ancient WGDs have been reported in Orobanchaceae (Yang *et al.*, 2015; Xu *et al.*, 2022) and *Cuscuta* (Convolvulaceae; Sun *et al.*, 2018). In Orobanchaceae, more than half of gene families containing haustoria-associated genes show evidence of ancestral gene duplications, which were interpreted as WGD-derived paralogues (Yang *et al.*, 2015). However, in both Orobanchaceae and *Cuscuta*, the WGDs most plausibly linked to haustorial evolution are shared with non-parasitic relatives. Furthermore, Yang *et al.* (2015) did not distinguish WGDs from alternative evolutionary hypotheses for how genes could become co-opted for haustorial function, including recent tandem duplications or pleiotropy. Although gene duplication is a well-recognized engine of evolutionary novelty (Ohno, 1970), plants tend to experience gene and genome duplications at high rates (Moore and Purugganan, 2005). Thus, although some haustorial genes might trace to WGD-derived neofunctionalization, additional evidence is needed to evaluate the broader range of evolutionary mechanisms by which genes are recruited for haustorial function following transitions to parasitism.

Over the past decade, newly developed genetic resources in Orobanchaceae have created the opportunity to investigate gene family evolution and tissue co-expression patterns of haustoria-associated genes within a comparative framework (Westwood *et al.*, 2012; Yang *et al.*, 2015; Xu *et al.*, 2022). Many of these studies (Westwood *et al.*, 2012; Yang *et al.*, 2015; Yoshida *et al.*, 2019) leveraged the same transcriptome datasets from the Parasitic Plant Genome Project (Westwood *et al.*, 2012) and contig-level genome assemblies. Advances in genome and transcriptome sequencing and assembly now allow for more accurate identification of haustoria-associated expression, orthology inference and reconstruction of gene regulatory modules, all of which will aid our ability to determine the identity and function of haustoria-associated genes and evaluate diverse evolutionary hypotheses for their origins.

In this study, we generated a new chromosome-scale reference genome and annotation of the facultative hemiparasite *Pedicularis groenlandica* Retz. (Orobanchaceae), which we use to investigate the evolutionary processes by which genes can be co-opted for haustorial function. *Pedicularis groenlandica* (Elephant's head lousewort) is an iconic flowering plant of western North American alpine meadows. It belongs to the globally distributed, facultative hemiparasitic genus *Pedicularis*, whose members form lateral haustoria and attach to hosts such as sedges, grasses and legumes (Piehl, 1963; Ren *et al.*, 2010; DiGiovanni *et al.*, 2017). *Pedicularis* is well studied for its distinctive pollination biology (Macior, 1983; Grant, 1994; Eaton *et al.*, 2012; Tong and Huang, 2016; Cohen *et al.*, 2025), but little attention has been paid to parasitism within the genus. Two chromosome-scale *Pedicularis* genome assemblies have been generated to investigate *Pedicularis* genome evolution, including investigations of gene loss, transposable elements and horizontal gene transfer (Xu *et al.*, 2022; Fang *et al.*, 2024).

With our newly generated genome and a transcriptomic time series of haustorial development, we aim to: (1) identify haustoria-associated genes in *P. groenlandica* and their overlap in expression with other tissues; (2) predict regulatory motifs associated with haustorial expression and their overlap with other tissues; (3) test the role of neofunctionalization in haustoria-associated gene evolution by comparing tissue-specific expression between paralogues that diverged before versus after the evolution of parasitism in Orobanchaceae; and (4) identify orthology among

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haustoria-associated genes in *P. groenlandica* and other parasitic plant lineages.

MATERIALS AND METHODS

Tissue collection and genome sequencing

Plant tissues for genome sequencing of *P. groenlandica* were collected at Rocky Mountain Biological Laboratory in Gothic, CO, USA in July 2024. Flow cytometry was performed on fresh leaf tissue to estimate genome size, and 5.0 g of leaf tissue was sent to Cold Spring Harbor Laboratory (Cold Spring Harbor, NY, USA) for high-molecular-weight DNA extraction, PacBio Hifi library preparation and sequencing on a PacBio Revio, and HiC library preparation and sequencing on an Illumina NextSeq2000 to generate 2 × 150 bp reads.

For genome annotation and differential expression studies, we collected three biological replicates of young leaf, stem, flower bud, pollen tube, root and mature haustoria tissues into RNAlater (Invitrogen), which was stored at −80 °C until extraction. To sample pollen tubes, we germinated fresh *P. groenlandica* pollen on pollen tube growth media, using the same methodology as Cohen *et al.* (2025). After 24 h, germinated pollen tubes were flash frozen over a dry ice–ethanol slurry and stored at −80 °C until extraction. RNA was extracted from all tissues using the Qiagen RNeasy Plant Minikit, and Illumina libraries were prepared and sequenced via Illumina MiSeq at Admera Health LLC (South Plainsfield, NJ, USA) to generate 2 × 150 bp reads.

Genome assembly and annotation

PacBio Hifi reads were filtered to retain only sequences >10 kb in length and were assembled into a phased contig-level assembly using Hifiasm v.0.19.8 run in 'HiC mode' (Cheng *et al.*, 2021). The primary contig-level assembly was analysed with the Arima Genomics Mapping Pipeline v.03 using YAHS for scaffolding (Zhou *et al.*, 2023), juicer (Durand *et al.*, 2016) to convert the scaffolded assembly into a Hi-C contact map, and JuiceBox Assembly Tools (JBAT; Durand *et al.*, 2016) for manual curation. We evaluated the completeness of our genome with BUSCO v.5.7.0 against eudicots_odb10.1 (Seppey *et al.*, 2019), then soft-masked the genome with RepeatModeler v.2.0.6 (Flynn *et al.*, 2020) followed by RepeatMasker v.4.1.7p1 (Smit *et al.*, 2013; Smit and Hubley, 2015).

We structurally annotated the genome using BRAKER3 v.3.0.8 with both RNA-sequencing (RNAseq) and protein evidence (Stanke *et al.*, 2006, 2008; Quinlan, 2014; Kovaka *et al.*, 2019; Pertea and Pertea, 2020; Brůna *et al.*, 2024; Gabriel *et al.*, 2024). We used RNAseq from all tissues and protein evidence from the Viridiplantae dataset in ORTHODB v.10 (Zdobnov *et al.*, 2021). RNAseq reads were trimmed of adapters and low-quality bases (sequences with phred quality score<Q30 retained) using TrimGalore v.0.6.10 (Krueger, 2015), and quality assessed using FastQC v.0.12.1 (Andrews, 2010). The annotation completeness was evaluated with BUSCO v.5.7.0 against the eudicots_odb10.1 dataset in transcriptome mode. We functionally annotated the genome using InterProScan with the –goterms tag, and identified the closest *Arabidopsis thaliana* orthologue of each gene with BLAST v.2.16.0 (Ye *et al.*, 2006). We conducted a synteny analysis in order to compare *P. groenlandica* with *Pedicularis cranolopha* and *Orobanche cumana* (Xu *et al.*, 2022) using MCScan (JCVItools; Tang *et al.*, 2024).

Haustorial time series tissue sampling and sequencing

To generate a time series of gene expression during haustorial growth and development, we sampled RNA from *P. groenlandica* grown in Petri dishes. We collected seeds from mature seed pods of *P. groenlandica* individuals in the same population as the genome plant. Seeds were removed from their pods, cleaned of excess tissue, and stored in dry envelopes at 4 °C until germination. *Pedicularis groenlandica* does not require a host for germination and early growth. We therefore germinated parasites and hosts separately, following the protocol of Westwood *et al.* (2012) for staged haustorial tissue collection in facultative hemiparasites.

To promote germination, seeds of *P. groenlandica* were treated with a 200 ppM solution of gibberellic acid for 15 min, then sterilized in a solution of 30 % bleach, 10 % ethanol, 60 % sterile de-ionized water and one drop of triton. We submerged the seeds in the sterilizing solution with agitation for 5 min, washed the seeds in sterile water for 3 min each, dried them completely, then germinated the dry seeds on plates with sterile 10 % Murashige-Skoog (MS) media. Seeds were placed ∼2 cm apart on each plate, and plates were wrapped in Parafilm and placed in a growth chamber on a diurnal cycle with 18 h of light at 22 °C and 6 h of dark at 18 °C.

We chose *Arabidopsis thaliana* as the host plant owing to its fast growth rate and because it has served as a successful host in previous *Pedicularis* haustorial assays (Kee Jia Yee, personal communication). Seeds of *A. thaliana* accession Col-0/Rédei-L209478*A* were purchased from Lehle seeds (Round Rock, TX, USA). Owing to their faster rate of germination, we began germinating *A. thaliana* seeds 3 days after *P. groenlandica*. We monitored seed germination over a 10 day period and selected only seedlings with both cotyledons open and primary root growth >2 cm for haustorial growth assays. To initiate the assay, we carefully removed *P. groenlandica* and *A. thaliana* seedlings, using sterilized forceps, and placed them together on a single, sterile, 10 % MS plate. We initially implemented trials in which the *A. thaliana* root was wrapped around the *P. groenlandica* root, but found that placing it ∼2 mm away from *P. groenlandica* promoted greater root and lateral haustorial growth towards *A. thaliana*, hence we proceeded to conduct all trials using this methodology.

Our assay included tissue sampled for RNA extraction at four distinct stages (1–4) to establish a time series of haustorial development and attachment (Fig. 1). We also sampled 1 cm of root tissue from *P. groenlandica* prior to host exposure (stage 0). At stage 1, we sampled 1 cm of root tissue 24 h after exposure to the host plant. At stage 2, bulbous 'swelling' haustoria were observed, which form prior to host penetration and attachment. At stage 3, haustoria have fully penetrated and attached to the host, which we confirmed by pulling lightly on the *A. thaliana* root; if the *P. groenlandica* root moved with it, we considered the

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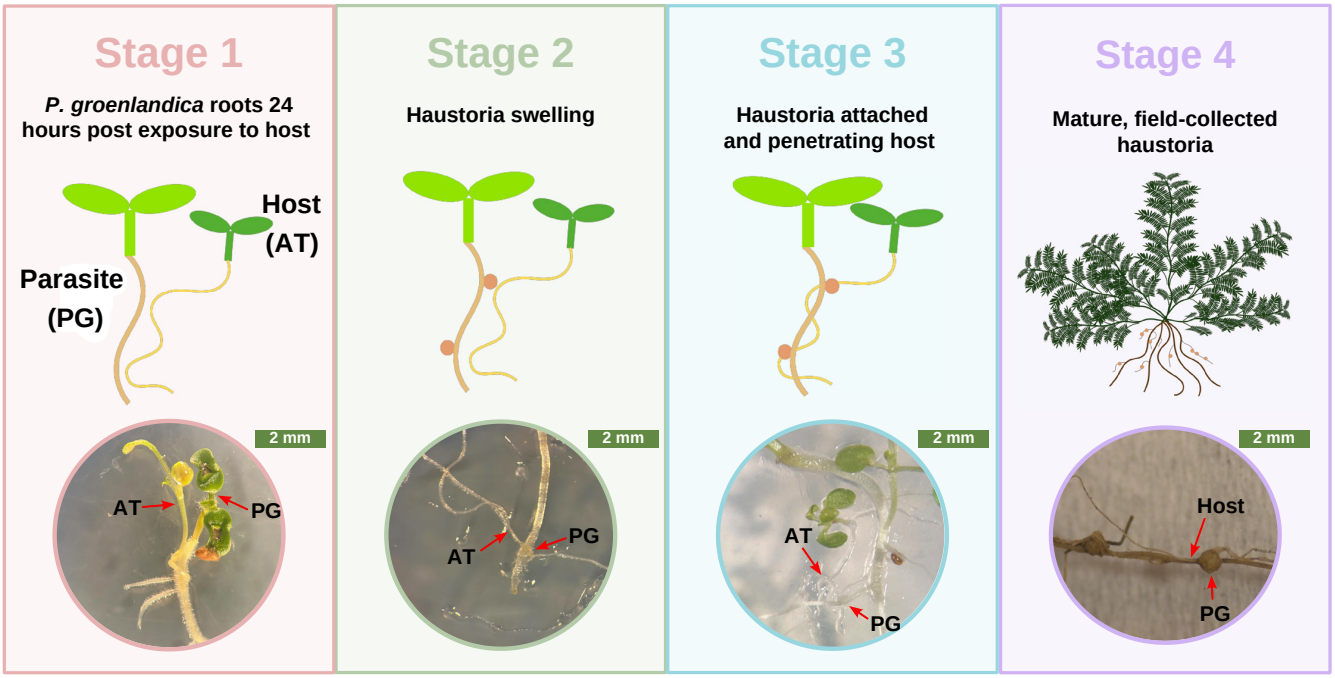


FIG. 1. Stages of haustorial development and attachment sampled in an assay involving the facultative hemiparasite *Pedicularis groenlandica* (PG). Stages 1–3 were sampled from haustoria attachment assays conducted in Petri dishes, using *Arabidopsis thaliana* (AT) as a host plant. For stage 4, we sampled mature haustoria in the field, where *P. groenlandica* was attached to a natural *Medicago* host (labelled 'Host'). Sampling stages were selected following Westwood *et al.* (2012).

haustoria fully attached. We also included mature, field-collected haustoria (stage 4) sampled from plants dug up during tissue collection for genome annotation. In stages 3 and 4, our samples included 1 mm of *P. groenlandica* root tissue above and below the haustorium. All tissues were collected in triplicate, immediately flash-frozen in liquid nitrogen, and stored at −80 °C until extraction. RNA was extracted using the Qiagen RNeasy Plant Minikit, and libraries were prepared using Poly-A selection and sequenced via Illumina MiSeq at Admera Health LLC (South Plainsfield, NJ, USA) to generate 2 × 150 bp reads.

Differential expression analysis and gene network co-expression analysis in haustorial tissues

RNAseq reads were trimmed of adapters and low-quality bases (<Q30) using TrimGalore v.0.6.10 (Krueger, 2015), with quality further assessed by FastQC v.0.12.1 (Andrews, 2010). All filtered reads, including reads from leaf, stem, root, flower bud and pollen tubes, were mapped to the final *P. groenlandica* genome using STAR v.2.7.11b (Dobin *et al.*, 2013). Read counts were obtained individually for each replicate at each stage of the haustorial assay (0–4) and for each non-parasitic tissue. Following alignment, we used featureCounts (Liao *et al.*, 2014) to count reads mapped to each annotated gene.

To identify differentially expressed genes, we analysed expression counts in the R package DESeq2 v.1.42.1 (Love *et al.*, 2014). We compared haustorial growth stages (stages 1–4) against non-parasitic tissue types (leaf, stem, root, flower bud, pollen tubes and stage 0) to identify genes that were either up- or downregulated in each haustorial time stage compared with non-parasitic tissue. To assess the expected similarity among samples within tissues visually, we performed a principal components analysis of the gene expression data. A gene was determined to be differentially expressed in haustoria if it had a log2FoldChange of >2 (upregulated) or <−2 (downregulated) relative to non-parasitic tissues, and an adjusted P-value < 0.05. We compiled a list of 'haustoria-associated' genes, which includes the set of genes that were differentially expressed in at least one stage (1–4) of the haustorial assay. We assessed the distribution of haustoria-associated genes across the *P. groenlandica* genome and calculated the correlation between haustoria-associated gene density and overall gene density using Pearson's correlation coefficient and Student's paired t-test.

To identify modules of co-expressed genes, we implemented weighted gene co-expression network analysis (WGCNA). Following the methodology of Gilman *et al.* (2022), we input the quantified transcripts per million (TPM) for each gene in each tissue to the WGCNA v.1.73 package in R (Langfelder and Horvath, 2008). We assessed which co-expression gene modules were correlated with haustorial tissue using Pearson's correlation coefficient and Student's asymptotic t-test. A module was designated as haustoria associated if its expression was significantly correlated with haustorial samples (stages 1–4) compared with non-parasitic/haustorial tissues (stage 0 and other tissues). To determine the functional descriptions of the genes in each module, we identified their *A. thaliana* homologues through BLAST (Ye *et al.*, 2006) using the TAIR *A. thaliana* Genome V (Lamesch *et al.*, 2012) as reference, then determined the Gene Ontology (GO) biological process terms associated with the *A. thaliana* homologues that were

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enriched in each module using the 'enrichGO' function in the *clusterProfiler* in R (v.4.10.1; Wu *et al.*, 2021).

Overlap of differentially expressed genes and regulatory motifs in haustoria and non-parasitic tissues

We investigated the overlap of haustoria-associated genes and regulatory motifs with leaf, root, stem, flower bud or pollen tubes. Identifying enriched regulatory motifs required initially grouping genes into co-expression modules with WGCNA (Langfelder and Horvath, 2008), then quantifying correlation with a particular tissue using Pearson's correlation coefficients, as described above for haustoria-associated modules. Thus, the labelling of both genes and modules as 'tissue-associated' involved pairwise comparisons (i.e. genes are upregulated in haustoria compared with non-parasitic tissues, or a module is correlated with haustoria compared with non-parasitic tissues). To avoid potential dilution of biologically relevant overlaps, haustoria and the tissue of interest were always compared against the same set of control tissues. For example, when investigating the haustoria–pollen tube gene overlap, we identified genes differentially expressed in haustoria relative to leaf, stem, root and flower bud, then intersected this with genes differentially expressed in pollen tubes relative to the same control tissues.

Following this approach, we identified genes that are differentially expressed in: (1) both haustoria and non-parasitic tissues; and (2) only haustoria, which we subsequently refer to as 'haustoria-specific' genes. Then, following the methods of Chomentowska *et al.* (2025), we searched for enriched motifs within 2 kb up- and downstream of genes that were both in WGCNA modules correlated with each tissue and differentially expressed in that tissue using the program XSTREME (Meme v.5.5.8; Bailey *et al.*, 2009). We identified overlapping enriched motifs between haustoria and leaf, root, stem, flower bud or pollen tubes by comparing their sequence and locus.

Because pollen tubes and haustoria had the highest overlap in both tissue-associated genes and regulatory motifs, we explored their functional properties further. For the set of genes that overlapped between haustoria and pollen tubes, we assigned molecular function GO terms as described above. We also functionally annotated the motifs by identifying transcription factor binding sites using TOMTOM (Meme v.5.5.8; Bailey *et al.*, 2009), comparing our enriched motifs against known motif annotations in the JASPAR database (Rauluseviciute *et al.*, 2024). Furthermore, we asked whether haustoria- and pollen tube-associated genes were grouped into co-expression modules more often than expected by chance. To do this, we constructed a contingency matrix that included the counts of overlapping haustoria and pollen tube-associated genes assigned to haustoria and pollen tube modules, respectively, as 'observed values'. To set null expectations, we performed a permutation-based test, in which we randomly shuffled module assignments while preserving the number of genes in each module. We conducted 2000 permutations and recalculated the overlap contingency table with these 'expected values'. We recorded the number of times that observed values were greater than or equal to expected values, then calculated empirical *P*-values. To account for multiple testing across all module pairs, *P*-values were adjusted using the Benjamini–Hochberg false discovery rate.

Analysis of haustorial gene trees and gene duplications

We evaluated the hypothesis that haustoria-associated genes evolved by neofunctionalization following gene or genome duplication events in a non-parasitic ancestor of Orobanchaceae. To do this, we downloaded protein sequences of four publicly available non-parasitic genomes from NCBI Genbank and JGI Phytozome [*Mimulus guttatus* (NONTOL v.4.0, DOE-JGI), *Salvia miltiorrhiza* (GCF_028751815.1; Pan *et al.*, 2023), *Solanum lycopersicum* (ITAG5.0; Zhou *et al.*, 2022) and *Arabidopsis thaliana* (TAIR10; Lamesch *et al.*, 2012)] and assembled orthogroups using Orthofinder v.3.0.8 (Emms and Kelly, 2019) run with default settings. We filtered orthogroups, retaining only those which contained at least one haustoria-associated gene and at least three genes in total. We cleaned and trimmed the multiple sequence alignments produced by Orthofinder using a custom Python script, then inferred gene trees with the LG+G model in RAXML-ng v.1.2.2 (Kozlov *et al.*, 2019) with support assessed from 100 non-parametric bootstraps. We used GeneRax v.2.1.3 (Morel *et al.*, 2020) to reconcile and root genes trees based on the species tree topology and the duplication–transfer–loss model of gene evolution.

We conducted tree-based operations in the Python package toytree v.3.0.10 (Eaton, 2020) to examine the phylogenetic relationships of haustoria-associated genes with respect to their paralogues and orthologues. Specifically, for each haustoria-associated gene, we examined the closest non-haustoria-associated paralogue and recorded whether it was either more closely related to the haustoria-associated gene than to the closest interspecific orthologue (i.e. it arose by duplication within Orobanchaceae, defined as a 'recent duplication') or equally or more distantly related to the haustoria-associated gene than the closest interspecific orthologue (i.e. it arose by duplication in an ancestor of Orobanchaceae, defined as an 'ancient duplication'). We also recorded whether there was no evidence of a duplication. If multiple paralogues were equally related to the haustoria-associated gene, one was chosen at random. For each gene, we recorded the tissue in which it was differentially expressed.

To validate our approach, we examined many gene trees visually, including several selected genes with known haustorial function. We also searched TAIR10 (Lamesch *et al.*, 2012) to identify the *A. thaliana* gene locus for homologues of genes that have been shown experimentally to affect pollen tube growth and signalling. The selected haustorial genes were: *ROOT MERISTEM GROWTH FACTOR* (*RGF*), which Fishman *et al.* (2025) demonstrated to be necessary for haustorial development in the hemiparasite *Phtheirospermum japonicum*; *KARRAKIN INSENSITIVE-2* (*KAI2*), which has been shown to have paralogues that enable ultra-sensitive strigalactone receptors in Orobanchaceae (Conn *et al.*, 2015; Wang *et al.*, 2021; Takei *et al.*, 2024); and two deep homologues of KAI2, DWARF14 (D14) and *DWARF14-LIKE2*

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(*DLK2*). The pollen tube growth and guidance genes were as follows: *Cyclic Nucleotide-Gated Channel 18* (*CNGC18*; von Besser *et al.*, 2006); *HAPLESS 2* (*HAP2*; Daher and Geitmann, 2012); and *Actin-Depolymerizing Factor 7* (*ADF7*; Gao *et al.*, 2016).

Modelling the effect of gene duplications

To quantify the relative importance of gene duplications to the evolution of haustorial specificity in gene expression, we compared the number of genes that are haustoria specific (i.e. only differentially expressed in haustoria) with those that are differentially expressed in multiple tissues including haustoria to ask whether haustoria-specific expression is predicted by the orthogroup having recent and/or ancient duplications. The number of *P. groenlandica* genes in each orthogroup was included as a covariate (log-transformed and centred before fitting). We fitted a population-averaged logistic model using generalized estimating equations (binomial, logit link) with genes clustered by orthogroup, an exchangeable working correlation and robust (sandwich) standard errors. Models were fitted using statsmodels (Seabold and Perktold, 2010) in Python. Two-sided Wald tests were used to assess coefficients, with odds ratios (ORs) and 95 % confidence intervals (CIs). We compared the effect of recent versus ancient paralogues via a Wald χ^2 test and computed average marginal effects.

Orthology and expression overlap of haustoria-associated genes among parasitic plant species

To investigate the extent to which haustoria-associated genes in *P. groenlandica* are shared with other parasitic plant lineages, we examined differential expression and orthology with other haustoria development datasets. We restricted our analyses to four datasets with transcriptomic time series of haustorial development analogous to our dataset (i.e. RNA sampled from at least stages 1–3); this included *Cuscuta campestris* (Bawin *et al.*, 2023), *Phtheirospermum japonicum* (Kokla *et al.*, 2021), *Striga hermonthica* (Westwood *et al.*, 2012) and *Phelipanche aegyptiaca* (syn. *Orobanche aegyptiaca*; Westwood *et al.*, 2012). The sequencing depth and sample replication in these datasets varied; *S. hermonthica* and *P. aegyptiaca* were generated in 2012, whereas *C. campestris* and *P. japonicum* were generated using more modern technologies since 2020. We used the 'control' samples of Kokla *et al.* (2022) that were not exposed to an exogenous nitrogen treatment.

In order to identify haustoria-associated genes by stage in each of these datasets, we followed the same pipeline that was used in *P. groenlandica*. A publicly available reference genome is available for each species (*C. campestris*, GCA_052055255.1; *Phtheirospermum japonicum*, GCA_014905375.1; *S. hermonthica*, GCA_902706635.1; and *P. aegyptiaca*, GWHBHPU00000000). We classified genes from *P. groenlandica* and the four published genomes into orthogroups using Orthofinder v.3.0.8 (Emms and Kelly, 2019). We quantified orthogroup overlap among haustoria-associated genes of species, then functionally annotated the haustorial orthologues by identifying their *A. thaliana* orthologues, followed by GO term enrichment analysis using clusterProfiler, as described above.

RESULTS

The P. groenlandica *genome*

The *P. groenlandica* assembled haploid genome size is 1.38 Gb, consistent with the flow cytometry estimate (C-value = 1.59; Supplementary Data Fig. S1). We scaffolded the genome into eight highly contiguous chromosomes, which include 91.6 % of nucleotide base pairs (Fig. 2). The BUSCO completeness score was 96.5 % (single-copy, 90.5 %; duplicates, 6.1 %). The *P. groenlandica* genome contained a high proportion of repetitive elements accounting for 76.48 % of the genome. A large proportion of repeats (44.14 %) are long terminal repeats. Our genome has high synteny to *P. cranolopha* (Xu *et al.*, 2022) but exhibits rampant rearrangement with respect to *O. cumana* var. *cumana*. We annotated 27 006 unique protein coding genes. The BUSCO score of the annotation was 94.3 % (single-copy, 59.6 %; duplicates, 34.7 %).

Haustoria-associated gene expression, function and regulation

Across our RNAseq libraries, we obtained an average of 27.84 million reads per replicate per tissue, of which a median of 20.89 million reads (83.01 %) uniquely mapped to the *P. groenlandica* genome. All protein coding genes were represented in the final count dataset. Differential expression analysis identified 5635 haustoria-associated genes across all four stages (Fig. 3A, B). The density of haustoria-associated genes across chromosomes was highly correlated with the density of all genes ($P < 0.001$; Fig. 3D).

In stage 1, 24 h after host exposure, 3734 genes were differentially expressed and showed enrichment for biological process GO terms relating to mitotic cell division, DNA replication and cell wall biogenesis. In stage 2, when haustoria began to swell, 2054 differentially expressed genes showed functions enriched in pigment biosynthesis, tetrapyrrole metabolism, ion transport and unidimensional cell growth. In stage 3, during host invasion, 679 genes were differentially expressed, which were enriched for pollen tube growth, pollination, cellular tip growth and unidimensional cell growth. In mature haustoria (stage 4), 2047 genes were differentially expressed, showing enrichment for light response and photosynthesis.

Of the 32 co-expression network modules identified by WGCNA, we identified ten that are significantly correlated ($P < 0.05$) with haustoria and assigned functional descriptions to each based on the most enriched GO terms among genes in the module (Fig. 3C). These co-expression networks generally correspond to functions that have previously been associated with haustorial development and attachment; they include: flavonoid biosynthesis (Albrecht *et al.*, 1999; Lozanova *et al.*, 2025), cell wall biogenesis (Cui *et al.*, 2016; Yoshida *et al.*, 2019), defence and immunity (Furuta *et al.*, 2021) and hormone signalling (Kirschner *et al.*, 2023). The association of modules with haustoria varied

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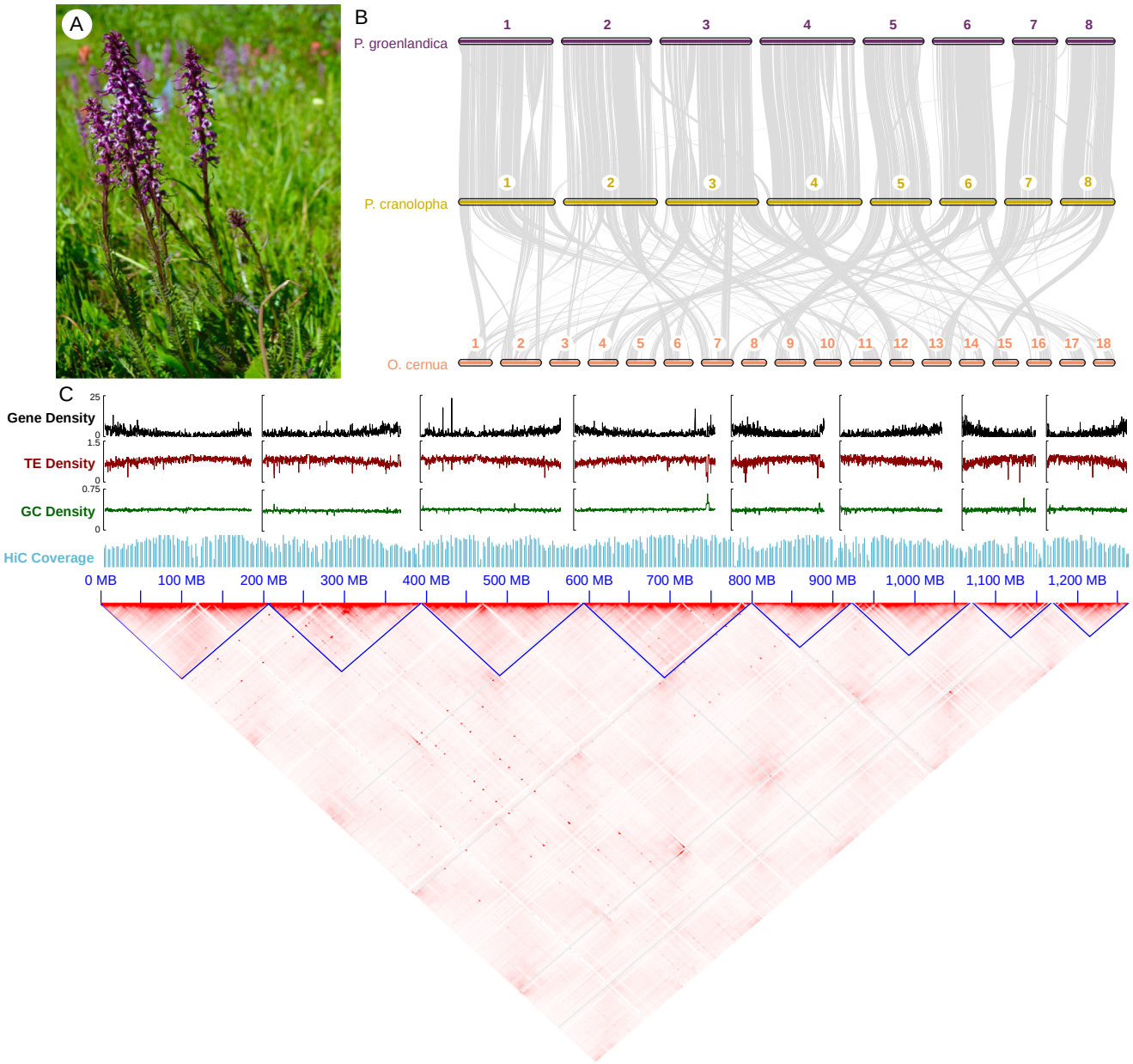


FIG. 2. Chromosome-scale assembly of the *Pedicularis groenlandica* genome. (A) Photograph of *Pedicularis groenlandica* (Elephant's head lousewort). (B) Gene synteny between the eight chromosomes of *P. groenlandica* and *Pedicularis cranolopha*. Both *Pedicularis* species have highly rearranged genomes with respect to the *Orobanche cumana* var. *cumana* genome (Xu *et al.*, 2022). (C) Gene density, transposable element (TE) density, GC content and HiC sequence coverage statistics along 10 kb sliding windows in the *P. groenlandica* genome. Genes and TEs are distributed evenly across the genome, and there are few peaks in GC content. We show that the HiC contact map shows eight contiguous chromosomes in *P. groenlandica* during correlation tests.

depending on the tissue types used for comparison. For example, when pollen tubes were excluded as a comparison tissue (i.e. we identified modules significantly correlated with haustoria in comparison to leaf, stem, root and flower bud), one of the haustoria-associated modules was enriched for pollen tube-associated functions (biological process GO terms such as pollination, unidimensional cell growth, pollen tube development and pollen tube growth, among others; Supplementary Data Fig. S2). Conservatively, we focus on the modules identified as significantly associated with haustoria in comparison to all other tissue types included in the study (Fig. 3C).

Haustoria-associated gene and regulatory motif overlap with non-parasitic tissues

Haustoria-associated genes have the highest overlap with pollen tubes, more than any other tissue type, including roots (Fig. 4A). Here, gene overlap means that the same loci were identified as differentially expressed in both haustoria and a particular tissue. Our enriched regulatory motifs analysis also showed the highest overlap between haustoria and pollen tubes (Fig. 4B), although the proportion of overlap was lower when measured by motif identity (Fig. 4A). Despite sharing a substantial set of genes, haustoria and pollen tubes often

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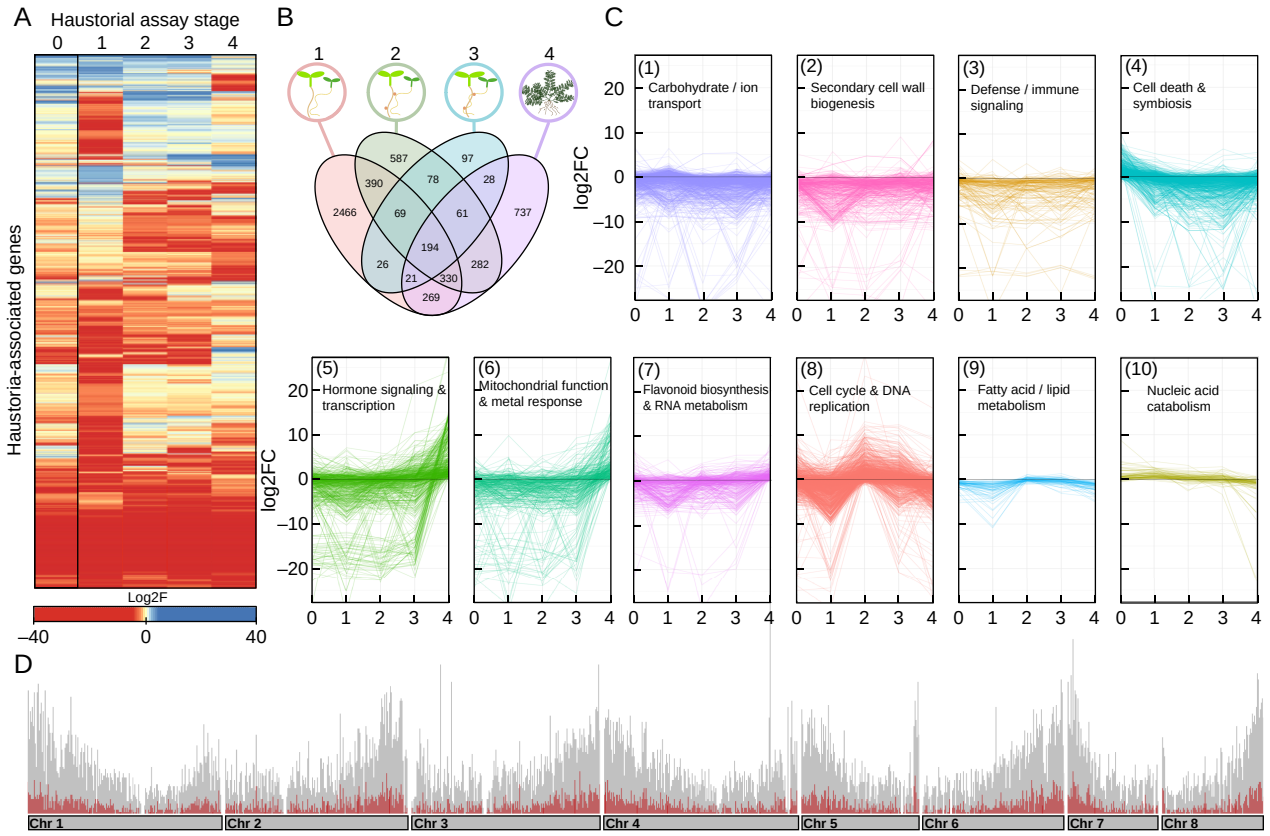


FIG. 3. Haustoria-associated genes and their gene expression in *Pedicularis groenlandica*. We identified 5635 haustoria-associated genes, including both up- and downregulated genes. (A) Heatmap of Log2FoldChange values for each haustoria-associated gene. Red indicates downregulation and blue upregulation. We include 'Stage 0', young roots pre-host exposure, to show changes in gene expression across the entirety of haustorial development and attachment. (B) Venn diagram showing the overlap in genes involved in haustorial response across the four stages. Stage 1, response 24 h post-host exposure, had the highest number of stage-specific genes. (C) Changes in gene expression among the sets of genes within ten haustoria-associated WGCNA modules, representing co-expressed molecular responses to host presence, haustorial initiation and haustorial attachment. Lines in each module plot show the change in gene expression of a single gene throughout the stages of haustorial development. Each module is labelled according to the most enriched GO biological process terms for the set of genes. (D) Bar chart depicting haustoria-associated gene density (red) compared with general gene density across the genome (grey), calculated in 10 kb sliding windows. The grey bars along the horizontal axis represent the eight chromosomes of *P. groenlandica*.

exhibit opposite trends in gene expression, whereby genes upregulated in pollen tubes are frequently downregulated in haustoria (Fig. 4B). This resulted in the expression profiles of haustoria and pollen tubes being the most distinct from one another among all tissues considered (Supplementary Data Fig. S3). Finally, haustoria and pollen tube-associated genes were grouped together into co-expression modules more often than we would expect owing to chance alone (Fig. 4C; false discovery rate-adjusted $P < 0.05$).

Divergence between haustoria-associated genes and their paralogues

The 5635 haustoria-associated genes identified in *P. groenlandica* were grouped into 3761 orthogroups, including alternative splice isoforms predicted by BRAKER3. After collapsing isoforms (which grouped into the same orthogroup 99.99 % of the time), our dataset contained 4718 *P. groenlandica* genes, corresponding to a mean ± s.d. of 1.25 ± 0.73 genes per orthogroup. We identified 2054 haustoria-associated genes (44 %) that have at least one non-haustoria-associated paralogue. Of these, 1211 (59 %) have a paralogue that arose from a duplication after the divergence of Orobanchaceae and *Mimulus*, whereas 1121 (55 %) have a paralogue that arose ancestral to this split. Only 278 (14 %) haustoria-associated genes have both recent and ancient paralogues in the same gene tree.

We examined the divergence and overlap in tissue-specific expression between paralogues in each category. For genes that were the product of recent duplications, paralogues of haustoria-associated genes were more frequently differentially expressed in pollen tubes (40 %) than any other tissue type (3–13 %), and the remaining half (52 %) were not differentially expressed in any non-parasitic tissue considered (Fig. 5A). For ancient paralogues, an even greater proportion are differentially expressed in pollen tubes (57 %) and very few in roots (19 %; Fig. 5B). In addition, paralogues expressed in pollen tubes are often found exclusively in this tissue, whereas root-associated paralogues tend to be expressed in multiple tissue types (Fig. 5A, B).

Selected gene trees

We detected haustoria-associated genes in both KAI2 and D14, which are strigolactone receptors in Orobanchaceae that function in host detection (Mutuku *et al.*, 2020).

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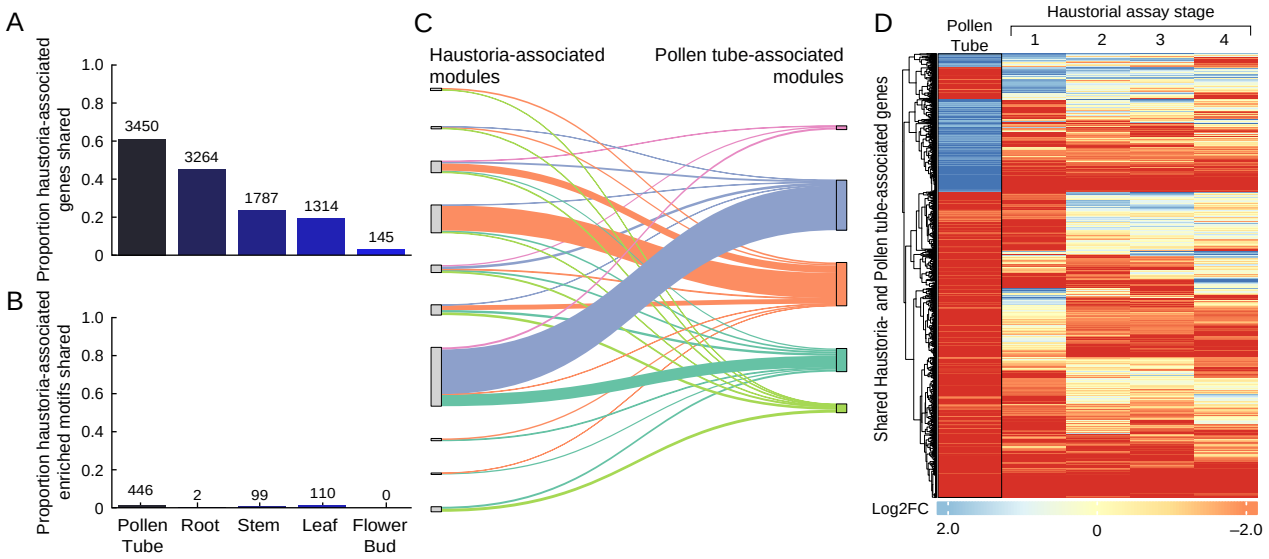


FIG. 4. Overlap of haustoria-associated genes and regulatory motifs with non-parasitic tissues. Bar charts showing (A) the proportion of haustoria-associated genes and (B) enriched regulatory motifs that overlap with non-parasitic tissues. We additionally report the count of overlapping genes and motifs above each bar. We show that haustoria-associated genes and motifs overlap most often with pollen tubes. (C) Relative grouping of genes into co-expression modules in haustoria and pollen tubes. Haustoria- and pollen-associated genes were grouped together in co-expression modules more often than expected owing to chance alone ($P < 0.05$ in each haustoria-associated module). (D) Heatmap of gene expression (Log2FoldChange) of overlapping haustoria- and pollen tube-associated genes. Blue values are upregulated genes and red are downregulated genes. Many shared genes show the opposite direction of expression in pollen tubes versus haustoria.

A haustoria-associated gene was also found within the homologue DLK2, which has not previously been implicated in parasitism. In all three orthogroups, the haustoria-associated genes were also differentially expressed in pollen tubes, in addition to roots in the case of DLK2.

Two homologues of ROOT GROWTH FACTOR that are associated with haustorial morphogenesis contain both haustoria-associated genes and non-haustoria-associated paralogues in *P. groenlandica* (Fig. 5G, H). In RGI4, an ancient duplication separates a haustoria-associated gene that is differentially expressed in several tissues, including during stage 1 of the haustorial assay, from a non-haustoria-associated paralogue that is uniquely expressed in pollen tubes. Likewise, RGFR2 contains an ancient duplication separating a haustoria-specific gene that was differentially expressed during stage 2 of the haustorial assay from a paralogue that was differentially expressed in pollen tubes and roots.

Three orthogroups were selected that contain *Arabidopsis* genes with known function in pollen-tube growth and guidance (CNGC18, HAP2 and ADF7; Fig. 5I–K). Each contained a haustoria-associated paralogue. In CNGC18 and ADF7, we observe a recent duplication, and the non-haustoria-associated paralogue is differentially expressed in pollen tubes and other tissues. In HAP2, the *P. groenlandica* orthologue is haustoria specific. A haustoria-associated gene in all three orthogroups was expressed during stage 1 of the haustorial assay. Overall, this set of tree visualizations (Fig. 5D–K) demonstrates the diversity of patterns of gene duplication and expression divergence across gene families containing haustoria-associated genes in *P. groenlandica*.

Modelling the effect of gene duplications

Using a generalized estimating equation logistic model, we found that the presence of a recent non-haustorial paralogue increases the odds that a haustoria-associated gene is haustoria specific (OR = 1.59, 95 % CI 1.27–2.00, $P < 0.001$). In contrast, ancient non-haustoria-associated paralogues showed no significant association (OR = 0.86, 95 % CI 0.70–1.05, $P = 0.13$). A direct Wald test confirmed the difference between recent and ancient effects ($\chi^2 = 23.19$, $P = 1.5 \times 10^{-6}$). The average marginal effects from the generalized estimating equation indicate that recent non-haustoria-associated paralogues increase the probability of haustoria-specific expression by 7.6 % (s.e. = 1.9, $P < 0.001$), whereas ancient paralogues are associated with a non-significant −2.5 % change (s.e. = 1.7, $P = 0.13$).

Diverging haustorial gene repertoires across parasitic plant lineages

Among the haustorial transcriptome datasets from five parasitic species (four Orobanchaceae and one *Cuscuta*), we identified a highly variable number of haustoria-associated genes. It was lowest among the two oldest datasets (*S. hermonthica* = 120, *P. aegyptiaca* = 513). In contrast, a much greater number of haustoria-associated genes was detected in the three recent datasets (*C. campestris* = 10 396, *P. japonicum* = 1058 and *P. groenlandica* = 5635; Fig. S5). We restricted our main comparative analysis to these three datasets (Fig. 6A**)**. Genes in shared orthogroups were expressed in each of the four stages of haustorial development (Fig. 6C). Orthology analysis identified haustoria-associated gene families shared between Orobanchaceae

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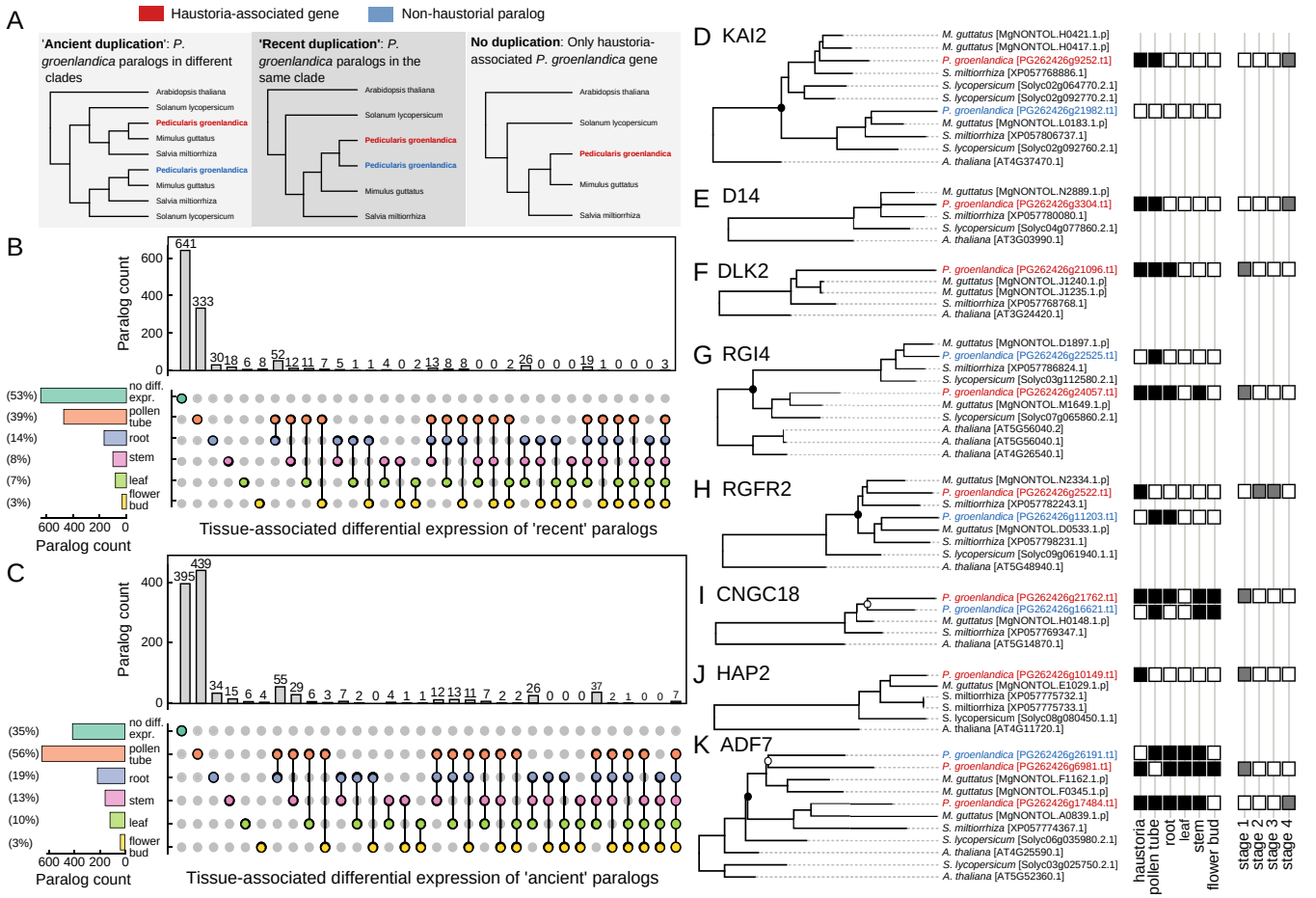


FIG. 5. Differential expression of *Pedicularis groenlandica* haustoria-associated genes and their paralogues across multiple plant tissues. (A) The closest non-haustoria-associated paralogue separated by either a 'recent' or 'ancient' gene duplication was identified in each orthogroup gene tree, if present. (B, C) The expression profiles of non-haustoria-associated paralogues are summarized in upset plots for 'recent' (B) and 'ancient' (C) paralogues, respectively. Most non-haustoria-associated paralogues of haustoria-associated genes are either not differentially expressed in any tissues or are differentially expressed in pollen tubes. (D–K) Example orthogroup gene trees showing expression profiles for haustoria-associated genes and their paralogues, if present. These orthogroups contain homologues associated with haustorial development or pollen tube guidance. Open circles indicate 'recent' gene duplications, and closed circles 'ancient' gene duplications. For haustoria-associated genes, we show the haustorial assay stage in which they were differentially expressed.

and *Cuscuta*, with an even greater number that are lineage specific (Fig. 6B). Within the five-species dataset, we identified only two orthogroups shared among all Orobanchaceae, in comparison to a majority of species-specific orthogroups (Supplementary Data Fig. S3). The highest orthogroup overlap was among the three recent datasets, with 660 shared between *P. groenlandica* and *C. campestris*, 258 shared between *P. japonicum* and *P. groenlandica*, and 90 shared between *C. campestris* and *P. japonicum*. Among all five species, the highest number of overlapping orthogroups occurred in those that contained genes that were differentially expressed during haustorial swelling (stage 2).

We evaluated the number of genes that made up the orthogroups in the three modern datasets. There were 600 haustoria-associated genes in the orthogroups that were shared by all three species: *P. groenlandica*, *P. japonicum* and *C. campestris* (Fig. 6B). These genes included those involved in tetrapyrrole and chlorophyll binding, aquaporins, glucosyltransferases, hydrolases and flavonoid-modifying enzymes (Fig. 6D). The lineage-specific orthogroups contained genes with distinct but complementary functions. In Orobanchaceae (overlap between *P. groenlandica* and *P. japonicum*), genes were enriched for functions related to cell wall modification, tropism and photosynthesis/oxidative stress. In holoparasitic *C. campestris*, genes were enriched for functions such as external encapsulating structure organization, cell wall organization and modification, pectin catabolic process and pattern specification.

DISCUSSION

Pedicularis groenlandica *genome*

Our *P. groenlandica* genome is a high-quality, chromosome-scale addition to the growing body of molecular data in the parasitic plant family Orobanchaceae. Including *P. groenlandica*, there are now 12 publicly available chromosome-scale genomes in the family (Ma *et al.*, 2021; Xu *et al.*, 2022; Chen *et al.*, 2024; Kim *et al.*, 2024; Twyford *et al.*, 2024; Bürger

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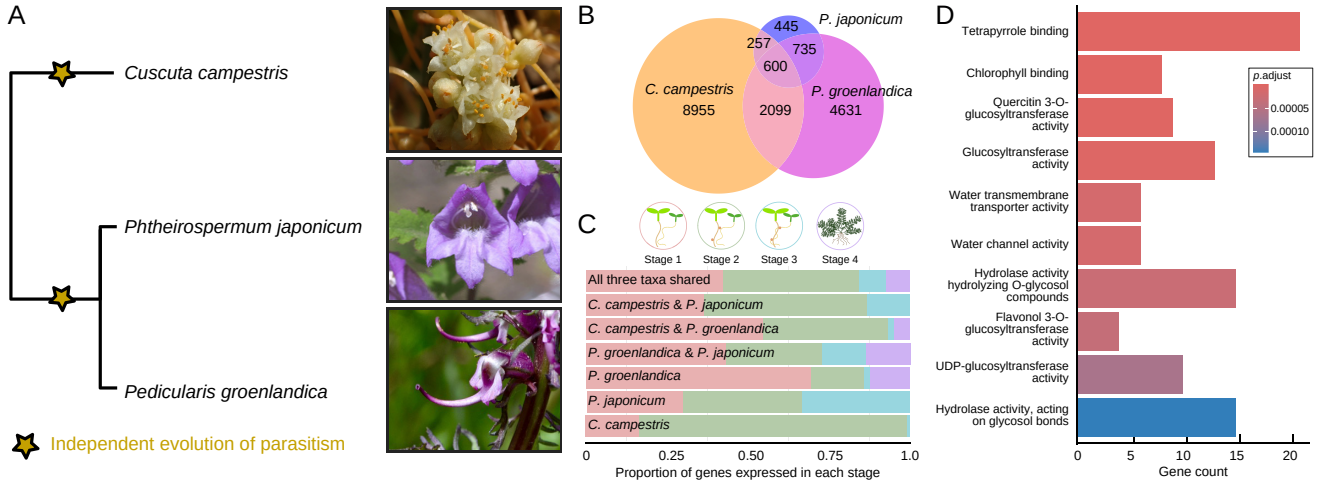


FIG. 6. Orthology of haustorial-associated genes in *Pedicularis groenlandica*, *Phtheirospermum japonicum* and *Cuscuta campestris*. (A) Cladogram showing the relationships between the three species, with two independent transitions to parasitism. Photographs of *C. campestris* and *P. japonicum* were taken from iNaturalist with permission (https://www.inaturalist.org/photos/62167474; https://www.inaturalist.org/observations/245056018). (B) Venn diagram of the overlap in orthogroups among the three species. The number in each section is the sum of genes in all three species that make up orthogroups in each category. (C) Bar chart depicts the breakdown of genes in each of the Venn diagram categories by the haustorial assay stage (1–4) in which they were differentially expressed in each species. *Cuscuta campestris* has a high number of species-specific stage 2 genes. (D) Molecular function GO enrichment analysis of the 600 genes in the orthogroups that overlap among all three species. The colour of bars shows the adjusted *P*-value associated with enrichment, with red being more enriched and blue less enriched.

et al., 2025; Zou *et al.*, 2026) and three in *Pedicularis* alone (Xu *et al.*, 2022; Fang *et al.*, 2024). HiC data support scaffolding the genome into eight chromosomes (Fig. 2D), which is consistent with our expectations based on available chromosome count data in *P. groenlandica* (Löve, 1982) and across *Pedicularis* (Index to Plant Chromosome Numbers chromosome reports), including two chromosome-scale genomes (Xu *et al.*, 2022; Fang *et al.*, 2024). We report relatively constant gene density and GC content across the eight chromosomes and high transposable element (TE) density (Fig. 2D).

The *P. groenlandica* genome is highly repetitive (76.5 % repeats), of which 44.5 % were annotated as TEs. Differential accumulation of TEs drives changes in genome size in plants, and in parasitic plants in particular (Neumann *et al.*, 2021). Parasitic plants might accumulate TEs disproportionately relative to non-parasitic plants because genes 'hitchhiking' on TEs can be transferred horizontally into parasitic plant genomes, driving genetic innovation, even in the face of increased rates of gene loss during transitions from autotrophy to heterotrophy (Cai, 2023). *Pedicularis groenlandica* has a high repeat content similar to other parasitic Orobanchaceae (*Phelipanche aegypitiaca* = 84.4 %, *O. cumana* = 79.8 % and *Orobanche coreulescens* = 86.3 %) and higher repeat content than non-parasitic Orobanchaceae (*Lindenbergia luchunensis* = 30.7 % and *Rehmannia glutinosa* = 67 %; Ma *et al.*, 2021; Kim *et al.*, 2024). The relationship of genome repeat content to parasitism warrants further exploration, particularly in a comparative context.

Xu *et al.* (2022) conducted a comparative genomic study of Orobanchaceae genomes, which included the *P. cranolopha* genome in some analyses. They observed limited synteny between *P. cranolopha* and two other Orobanchaceae genomes (*L. luchunensis* and *O. cumana* var. *cumana*). Consistent with the findings by Xu *et al.* (2022), the eight chromosomes of *P. groenlandica* show clear synteny with *P. cranolopha*, with some small inversions and translocations, but a high amount of rearrangement with respect to *O. cumana* (Fig. 2B). From this, we conclude that *Pedicularis* genomes have undergone a major reorganization since splitting from their common Orobanchaceae ancestor.

Haustorial gene function and regulation in P. groenlandica

Haustoria-associated genes make up 20.65 % of all protein coding genes in *P. groenlandica*, indicating that haustorial development and attachment leverages a wide range of molecular processes and elicits a genome-wide response. Functions of genes across the haustorial developmental stages (Fig. 4B) were consistent with the functional categories we expected based on previous haustorial RNAseq studies. For example, during stage 1, we observed the differential expression of genes involved in cell division and cell wall expansion, which is likely to be related to the initiation of haustorial growth (Ishida *et al.*, 2016; Cui *et al.*, 2020). Stages 3 and 4 had similar expression patterns to each other, showing enrichment for cellular tip growth-related genes. Cui *et al.* (2016) found that cellular tip growth and cell elongation are major mechanisms by which haustoria and functional haustorial hairs form in *Phtheirospermum japonicum*, a facultative hemiparasite that is closely related to *Pedicularis* within Orobanchaceae.

Likewise, the functional associations of haustoria-associated gene co-expression networks included biological processes with known relevance to haustorial growth and invasion (e.g. flavonoid biosynthesis; Fig. 3C). Flavonoids are known to be related to haustoria formation in *Triphysaria versicolor* (Albrecht *et al.*, 1999) and in the genus *Cuscuta* (Lozanova *et al.*, 2025). Additionally, we identified a module of genes related to defence and immune response

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(Fig. 3C), which we expect to include genes interacting with host defence pathways during parasitic invasion. Hormone signalling and ion transport might also relate to host–parasite cross-talk, and we saw enrichment of genes involved in auxin, ethylene, salicylic acid and phenylpropanoids, each of which has been linked to host–parasite signalling in other parasitic plants (Ishida *et al.*, 2016; Cui *et al.*, 2020; Wakatake *et al.*, 2020; Mutuku *et al.*, 2021). Our results thus tell a consistent functional story about haustorial development compared with other parasitic lineages in which haustoria-associated genes have been identified.

Pollen tubes and the evolution of haustoria-associated genes

Both Yang *et al.* (2015) and Yoshida *et al.* (2019) found that orthologues of haustoria-associated genes in *T. versicolor*, *Phelipanche aegypotiaca*, *S. hermonthica* and *Striga asiatica* were enriched in pollen-related functions in *A. thaliana*. We hypothesized that this pattern results from the overlap in the function of haustoria and the process of pollen tube growth, which, like haustorial invasion, requires signal detection, growth through foreign tissue and directed unidimensional cell growth. In our study, 61 % (3450 of 5635) of haustoria-associated genes were differentially expressed in both haustoria and pollen tubes, which represented a significant overlap (Fischer's exact test, $P < 10^{-70}$). This overlap was greater than any other non-parasitic tissue considered, including roots (Fig. 4A). The extent of the haustoria–pollen tube relationship was surprising, because haustoria-associated genes have previously been hypothesized to be co-opted from predominantly root-associated genetic pathways.

Pollen tube growth is known to require ethylene and auxin signalling (Guan *et al.*, 2013; Chen *et al.*, 2014; Weijers and Wagner, 2016; Cui *et al.*, 2020; Mou *et al.*, 2020), in addition to genes in the expansin superfamily (Cosgrove, 1997; Sampedro and Cosgrove, 2005), all of which are also correlated with haustoria development (Yoshida *et al.*, 2016, 2019). In addition to high overlap in differentially expressed genes among haustoria and pollen tubes, the two tissues also share the highest number of enriched regulatory motifs. Among the shared haustoria and pollen tube motifs, the most common transcription factor family with which they interact is ERF/DREB (Supplementary Data Fig. S4B), which regulates ethylene signalling. Yoshida *et al.* (2016) argue that the co-option of the lateral root development programme led to the evolution of haustoria across the 12 parasitic plant lineages, focusing in particular on pathways involved in auxin biosynthesis and regulation. Across our 5635 *P. groenlandica* haustoria-associated genes, 155 were involved in known auxin pathways, and of those, 62 auxin genes were additionally associated with pollen tubes, 18 were associated with roots, and 33 were associated with all three tissues. Despite known functioning of auxin pathways in roots, our results suggest that, in *P. groenlandica*, auxin functionality in haustoria has predominantly been co-opted from pollen tube-associated auxin pathways.

One explanation for the high proportion of haustoria-associated genes that were differentially expressed in pollen tubes is that genes function pleiotropically in the two tissues. We expect pleiotropy to give rise to divergence in *cis*-regulatory elements that mediate gene expression, which is supported by our observation that haustoria and non-parasitic tissues share many of the same genes but few enriched regulatory motifs (Fig. 4B). Many genes expressed in both haustoria and pollen tubes were similarly grouped into co-expression modules (Fig. 4C) despite frequently exhibiting divergent directions of gene expression (Fig. 4D). These findings are consistent with pleiotropy. Specifically, divergence in tissue-specific regulation might have been favoured after the transition to parasitism owing to antagonistic pleiotropy, whereby mutations in a gene that are beneficial in one tissue might be deleterious in another (Hoekstra and Coyne, 2007). The evolution of tissue-specific *cis*-regulatory elements has been shown to alleviate the burden of antagonistic pleiotropy for *Pitx1* expression in sticklebacks (Chan *et al.*, 2010), and variable patterns in expression reduce deleterious effects of sexually antagonistic genes in a number of systems (Gompel *et al.*, 2005; Mank *et al.*, 2008, Wright *et al.*, 2018). Together, these results suggest that divergence in the regulation of pollen tube-associated genes might underlie much of the novel gene expression in haustoria.

A majority of the 5635 haustoria-associated genes in *P. groenlandica* showed evidence of association with pollen tubes, either because they were differentially expressed in pollen tubes or nested within pollen tube-associated gene families, or both. Among haustoria-associated genes without a *P. groenlandica* paralogue in their orthogroup, 66 % (1763 of 2664) were themselves differentially expressed in pollen tubes; among those nested in orthogroups that contain a paralogue, that paralogue was differentially expressed in pollen tubes 52 % of the time (1061 of 2054). These patterns suggest that haustoria have disproportionately co-opted functions from pollen tubes, and that pollen tube-associated genes and gene families have been integral to the evolution of haustoria-associated genes in *Pedicularis*. Although shared function cannot be inferred from expression alone, our findings provide strong support for a close relationship between these two tissue types that future studies should address through more direct experimentation.

Alternative explanations for the co-option of haustoria-associated genes

Yang *et al.* (2015) argue that haustoria-associated genes evolve through regulatory neofunctionalization following ancient gene duplications, which they correlate with ancestral WGDs. If haustoria-associated genes were derived from WGDs, we would expect to identify ancient duplications in the histories of haustoria-associated genes, in addition to likely haustoria-specific gene expression. In *P. groenlandica*, only 19.2 % of haustoria-associated genes showed evidence of diverging from non-haustoria-associated paralogues following an ancient duplication. Furthermore, from examination of gene tree topologies alone, the methodology used by both Yang *et al.* (2015) and the present study, whether ancient duplications were derived from a WGD or tandem duplication that occurred in a shared common ancestor cannot be distinguished. We show that the evolution of haustoria-specific expression is significantly more likely following a recent duplication and is not affected by the

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presence of ancient duplications. This is consistent with the idea that recent, rather than ancient, duplications have facilitated neo- or subfunctionalization of haustoria-specific genes. The predictive effect, however, is small, because the presence of a recent paralogue makes it only 7.6 % more likely that a gene will exhibit haustoria-specific gene expression.

Both regulatory neofunctionalization and pleiotropy are contexts in which new phenotypes can evolve through variation in gene expression. Within our gene tree dataset, we observed topologies and differential expression patterns in haustoria-associated and non-haustoria-associated paralogues consistent with three, non-mutually exclusive, evolutionary explanations for the advent of haustorial function: (1) pleiotropy; (2) neofunctionalization following a recent duplication; and (3) neofunctionalization following an ancient duplication. We hypothesize that co-option of non-parasitic functions for haustorial development might occur initially via pleiotropy, and that subsequent recent (potentially lineage-specific) gene duplications have given rise to paralogues that diverge in their functions and tissue-specific expression patterns, perhaps to alleviate the burden of pleiotropy. Fishman *et al.* (2025) provide evidence of neofunctionalization following a recent, segmental tandem duplication in RGF orthologues in *Phtheirospermum japonicum*, consistent with our hypothesis for haustoria-associated gene evolution. Further studies that validate haustoria-associated gene functions in non-parasitic tissues and compare patterns of selection acting on genes that are uniquely and non-uniquely expressed in haustoria will be needed to investigate this hypothesis comprehensively.

Divergent pathways towards the evolution of haustoria

We expected to identify a core set of genes across Orobanchaceae that contribute to haustorial function in the family, derived in the most recent common ancestor of Orobanchaceae. Although there is some overlap among haustoria-associated genes across the parasitic plant lineages considered, a high proportion of haustoria-associated genes are species specific (Fig. 6). Likewise, Yang *et al.* (2015) found that most haustoria-associated genes were species specific when comparing *T. versicolor*, *S. hermonthica* and *P. aegyptiaca*. Their results suggest that hemiparasitic versus holoparasitic haustorial function necessitates the recruitment of diverse sets of genes, with relatively small overlap.

Even between the two closely related facultative hemiparasites *P. groenlandica* and *P. japonicum*, a high proportion of haustoria-associated genes were species specific. This result is particularly interesting in light of our findings that: (1) *Pedicularis* have highly rearranged genomes with respect to other Orobanchaceae; and (2) recent duplications predict whether a haustoria-associated paralogue is differentially expressed only in haustoria. Together, these results suggest that since the evolution of haustoria in the common ancestor of Orobanchaceae, a substantial amount of variation has evolved among descendant lineages in the recruitment of genes into haustoria-associated functions. Many aspects of the biology of these organisms have probably impacted this process. For example, *Pedicularis* has experienced an adaptive radiation since the Miocene, where it now comprises >600 species distributed mostly throughout montane and arctic ecosystems (Eaton and Ree, 2013). Whether *Pedicularis* has evolved variation in haustoria-associated genes faster than other parasitic lineages will require further comparative analyses, including examination of the distribution of haustoria-associated genes with respect to synteny. The closely related genus *Castilleja* offers an opportunity for comparison with a similarly large and diverse clade (Li *et al.*, 2019).

Taking advantage of haustoria RNAseq data from *C. campestris* (Bawin *et al.*, 2022, 2023), we performed the first cross-family comparison of haustoria-associated genes in Orobanchaceae and *Cuscuta*. Interestingly, haustoria-associated genes shared among all three lineages are frequently involved in water transfer functions. The hemiparasitic *P. groenlandica* and *P. japonicum* form haustoria that access only host xylem (Yoshida *et al.*, 2016), whereas the haustoria of holoparasitic *C. campestris* penetrate both the xylem and phloem (Yoshida *et al.*, 2016). The overlap in genes that function in tetrapyrrole binding, glucosyltransferase and water transmembrane/channel molecular pathways in both lineages (Fig. 6D) suggests potential commonality in the way that hemi- and holoparasites invade xylem and transfer water from their hosts. This is consistent with the hypothesis that holoparasitism evolves subsequent to the evolution of hemiparasitism, and that the first haustoria in parasitic lineages probably resembled the lateral haustoria of hemiparasites (Westwood *et al.*, 2010).

Conclusions

This study presents strong evidence for the importance of pollen tube-associated genetic pathways in the evolution of haustoria in *P. groenlandica*. We highlight the need to evaluate diverse mechanistic explanations for the co-option of genes for haustorial function in parasitic plants and, more broadly, the ways in which changes in gene expression can contribute to the evolution of novel phenotypes. Modern genome assemblies and annotations have greatly improved our ability to identify and compare genes across parasitic plant lineages, and we underscore the similar importance of high-quality haustorial transcriptomic datasets, which provide a powerful comparative framework that has revealed the variability in haustorial gene evolution both within Orobanchaceae and across parasitic plant lineages.

SUPPLEMENTARY DATA

Supplementary data are available at *Annals of Botany* online and consist of the following. Figure S4: flow-cytometry report for *Pedicularis groenlandica* (PG) compared with pea. Figure S2: haustoria-associated WGCNA modules when haustorial tissues are compared with all non-parasitic tissues except pollen tubes. We show the change in gene expression (Log2FoldChange) for genes in each module across the four stages of our haustorial growth assay. When pollen tubes are excluded, we find one module with functions specifically related to pollen tube growth.

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Figure S3: principal component analysis (PCA) based on differential expression of each tissue, calculated in DESeq2. We show that pollen tubes (bottom left) have distinct differential expression profiles with respect to other tissues, in terms of up- versus downregulation. All other tissues group as expected, with haustoria and roots, in addition to leaves and stems grouping together. Figure S4: functional annotations of overlapping haustoria- and pollen-tube-associated genes and enriched motifs. (A) GO term molecular function enrichment analysis of the 3450 overlapping haustoria-associated and pollen tube-associated genes (PT). Colour of the circles shows the adjusted *P*-value (lighter colour/lower value are more significantly enriched). Dot area indicates the number of genes in that GO molecular function category. (B) Bar chart showing the transcription factor (TF) families to which overlapping haustoria- and pollen tube-associated regulatory motifs correspond. Figure S5: orthology analysis of haustoria-associated genes across five parasitic plant lineages. (A) Cladogram of the five species included. Four (*S. hermonthica*, *Orobanche aegyptiaca*, *Ptheirospermum japonicum* and *Pedicularis groenlandica*) are in Orobanchaceae, whereas *Cuscuta campestris* represents an independent transition to parasitism. (B) Venn diagram showing the overlap in orthogroups that contain haustoria-associated genes among the five species. We identified more genes in the three modern datasets (*P. japonicum*, *P. groenlandica* and *C. campestris*), and they exhibit much higher orthogroup overlap with each other than *S. hermonthica* and *O. aegyptiaca*.

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DATA AVAILABILITY

Herbarium vouchers for *P. groenlandica* were deposited at Rocky Mountain Biological Laboratory (RMBL) herbarium (voucher RMBL0007650) and New York Botanical Garden William and Lynda Steere Herbarium. Owing to a backlog of vouchers, our specimens had not yet been processed at the time of publication. They will be able to be found using the collector's and species names (R. Cohen) after they have been fully processed by New York Botanical Garden.

All genomic and transcriptomic data, including the reference genome, can be found on NCBI under BioProject PRJNA1461970. The haustoria-associated gene orthology dataset and functional annotation of the *P. groenlandica* genome are accessible on Dryad at https://doi.org/10.5061/dryad.w3r228174.

COMPETING INTERESTS

We declare no competing interests.

AUTHOR CONTRIBUTIONS

R.O.C. and D.E. contributed equally to study design. R.O.C. collected all data and conducted all differential expression analyses. R.O.C. and A.C. did gene co-expression network and motif enrichment analyses. R.O.C. and D.E. conducted the gene duplication-related analyses. R.O.C., D.E., A.C. and L.C. contributed to the writing of the manuscript.

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